MMCRD: An Effective Algorithm for Deploying Monitoring Point on Social Network

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Abstract. Complex relationships and restrictions on social networking sites are severe issues in social network data acquisition. Covering information of all users in social network and ensuring timeliness of data acquisition is of great significance. Therefore, it is critical to develop an efficient data acquisition strategy. In particular, smart deployment of monitoring points on social networks has a great impact on data acquisition efficiency. In this paper, we formulate the monitoring point deployment issue as a capacitated set cover problem (CSCP) and present a maximum monitoring contribution rate deployment algorithm (MMCRD). We further compare the proposed algorithm with random approximation deployment algorithm (RD) and maximum out-degree approximation deployment algorithm (MOD), using synthetic BA scale-free networks and real-world social network datasets derived from Facebook, Twitter and Weibo. The results show that our MMCRD algorithm is superior to the other two deployment algorithms, since our approach can monitor the entire social network users by monitoring at most 12% of users, and meanwhile, guarantee timeliness.

Keywords: Data acquisition · Timeliness
Monitoring point deployment · Capacitated set cover problem
MMCRD

1 Introduction

Social networking sites (e.g., Facebook, Twitter and Weibo) have emerged as main channel for people to share and exchange information on a wide range of real-world events. These events range from popular and well-known ones (e.g., a concert of a famous music band) to a small-scale local event (e.g., a local gathering and protest). Short messages posted on social networking sites reflect events that are taking place and spread in the form of "nuclear fission" through relationship network. Monitoring real-time messages in social network has an important significance in burst event detection [1], rumor detection [2], natural disaster detection [3], event topic identification [4], stock markets forecasting [5] and other fields.

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Three technologies are commonly adopted to acquire social networking data, namely application programming interface (API), asynchronous loading based on communication simulation and reverse crack engineering. API is restricted by resources and has strict limits on the frequency, authority and content of call interfaces. Asynchronous loading need to overcome a variety of measures against crawler and analyze the perplexing dependencies among requests, which is difficult to acquire complete data. Reverse crack engineering requires a huge and unmanageable reverse engineering, the escalating security measures make the cost of crack engineering increase and the risk is difficult to control.

This paper use API to acquire social networking data. Two key requirements of acquiring social networking data are a full coverage of users in network and timely data access as data are generated on the networks continuously.

To meet the first requirement, in traditional ways, we need to invoke API periodically for each user on social network, which will certainly require massive API calls due to the sheer number of users. To meet the second requirement, we require to monitor users as less as possible. Obviously, with given amount of resources, we have to develop an effective strategy to deploy monitoring points on social network.

Based on the analysis of user relationships and information dissemination in social network, we formulate monitoring point deployment issue as a CSCP and propose MMCRD. We further compare the proposed algorithm with RD and MOD in terms of the relationship between monitor capacity and required number of monitors by using real-world datasets derived from Facebook, Twitter and Weibo and synthetic BA scale-free network. The results show MMCRD is superior to the other two deployment algorithms, we only need to monitor at most 12% users in order to achieve full coverage and guarantee timeliness by using MMCRD.

2 Related Work

Online social network is a social structure which is formed by the individuals and relationships between the individuals. The relationships between individuals can be friends, relatives, behavior interactions and other relationships. Active behaviors of individuals in social network promote the constant changes of network structure, which makes social network hold the characteristics of community interaction [6], information dissemination [7] and community evolution [8]. Social networks have a large volume of valuable information. Compared to crawl news sites, blogs, forums and peer-to-peer systems, acquiring social networking data is really a difficult job. Social networks consist of three types of data, namely user profiles, user relationships and user-generated contents (tweets). User profiles and user relationships are relatively stable while user-generated contents change constantly.

In recent years, a great deal of research work acquire social networking data by means of traversing user graph which is feasible in most social networking sites, including Facebook [9,10], Ebay [11,12], Youtube [13,14], Flickr [15,16],

Twitter [17,18] and Weibo [19]. User graph crawling is an iteration process that starts from one or more seed users and depends on user relationships to discover new user in each iteration. It can exploit the skeleton of social network and easy to acquire user profiles and relationships. In addition, several strategies are commonly adopted to traverse user graph, including Bread-First-Search, Depth-First-Search, Forest Fire, Snowball Sampling and Respondent-Driven Sampling, they are differ in order through which they visit new users in each iteration.

Crawling is the most frequently used solution for data acquisition and is effective indeed to obtain user profiles and user relationships whereas is not suitable for acquiring user-generated contents. In most cases, acquiring user-generated contents is based on acquired users that means we can not satisfy the requirement of full coverage as long as there is no acquisition for some users. Behind the difficulty mentioned above is mainly because of (1) huge size of user relationship graph. (2) restrictions on social networking sites, e.g., IP banning.

Due to the above-mentioned problems, it is real a challenge to acquire the constantly update of user-generated contents timely. The purpose of this paper is to propose an effective monitoring point deployment algorithm to achieve full coverage of users and timely acquisition of user-generated contents.

3 Problem Formulation

3.1 Deploy Monitoring Points on Social Network

Four main behaviors are included in social network, namely *follow*, *post*, *retweet* and *mention*. The *follow* behavior builds a following relationship which require no mutual approval and is often regarded as a form of subscription. Other behaviors (*post*, *retweet* and *mention*) are relevant to user-generated contents. The relationships among users are built based on *follow* behaviors. If user A follows user B, user A is called a follower of user B, conversely, user B is called a friend of user A. In such a situation, if user B posts a new tweet (or retweet), this new tweet (or retweet) will appear in the main page of user A. In other words, if user A follow user B, information generated by user B will diffuse to user A. The following relationships and information diffusion can be shown as Fig. 1. The solid lines denote following relationships and the dash lines denote information diffusion directions. For example, user A follow user B and user C, the information generated from user B and user C will disseminate to user A. Therefore, as long as we deploy a monitor on user A, we can obtain all the information generated from user A, B and C. Other users are the same situation. Therefore, if we put monitors on user A, D, E, F, we can acquire the full information generated from all users in Fig. 1.

Based on user following relationships and information diffusion process, we can deploy monitoring points on a set of users: $U = \{u_1, u_2, ..., u_m\}$, and let each user u_i to cover a group of users, for example, the group of users covered by u_i can be represented as $V_i = \{v_{i1}, v_{i2}, ..., v_{ij}\}$. If

$$\bigcup_{i=1}^{m} V_i = \Omega \tag{1}$$

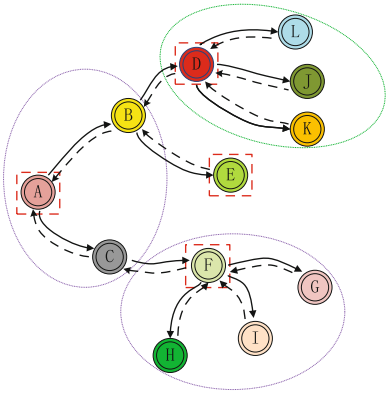


Fig. 1. User following relationships and information diffusion (The solid lines denote user following relationships; The dash lines denote information diffusion directions).

where Ω represents the whole users on a social network. Then we can monitor the set of users U to achieve full coverage of the users in social network.

Definition 1. We use a directed Graph $G = (V, E)$ to represent a social network compose of users and user relationships. Let $V = \{v_1, v_2, ..., v_n\}$ represent users in social network and $E = \{e_{1,1}, e_{1,2}, ..., e_{n,n-1}\}$ represent user relationships. For example, if exists a relationship between user v_i and user v_j, it represents that user v_i follow user v_j and also represents that $e_{i,j}$ contains in set E. Based on following relationships, if exists a subset of vertices $V' \subseteq V$, for each user $v_i \in V'$, we can obtain its following set which can be represented as F_i, Let S_i represent v_i combining with F_i. If any user belong to V, it can be covered by at least one subset S_i, then we call V' is the monitor set of graph G. If the number of elements of V' is minimum, V' is called a minimum set cover of graph G.

In this way, we formulate monitoring point deployment problem as a capacitated set cover problem. Next we will discuss how to determine appropriate monitoring points to achieve full coverage of the users in network and guarantee timeliness.

3.2 Set Cover Problem Formulation

Assume we have a set $U = \{v_1, v_2, ..., v_n\}$ and m subsets $S = \{S_1, S_2, ..., S_m\}$, where $S_j \subseteq U$, $j \in J$ and $J \subseteq \{1, 2, ..., m\}$. If the following equation can be satisfied,

$$\bigcup_{j \in J}^{m} S_j = U \tag{2}$$

where we call $\bigcup_{j \in J}^{m} S_j$ the set cover of U. A set cover the problem of selecting a collection of subsets satisfied that every element in U belongs to at least one of

the subset $S_j(j \in J)$. The minimum set cover optimization problem is to find a set cover that uses the fewest subsets.

We can transform the monitoring point deployment problem into a minimum set cover problem, which aims to achieve the full coverage with minimum subsets. We introduce a decision variable x_j to each subset $S_j(j = 1, 2, ..., m)$, which is described as below.

$$x_j = \begin{cases} 0 & if\, j \in J \\ 1 & if\, j \notin J \end{cases} \tag{3}$$

The minimum set cover problem can be formulated as the following integer program.

$$\begin{aligned} \min\ & z = \sum_{i=1}^{m} x_i \\ s.t.\ & \sum_{j:e_i \in S_j} x_j \geq 1 \quad i = 1, 2, ..., n \\ & x_j = 0, 1 \quad i = 1, 2, ..., n \end{aligned} \tag{4}$$

The constrain $\sum_{j:e_i \in S_j} x_j \geq 1$ ensures each element e_i of S can be covered by a subset S_j.

In addition to the full coverage, monitoring point deployment also require timeliness. The delay of data acquisition is determined by the number of elements covered by each vertex. Due to information dissemination among users in the form of tweet, we need to consider tweet frequency in data acquisition. Assume the maximum tweet frequency is f tweets per second, data acquisition rate is p tweets per second and tolerable delay is t seconds, the number of elements in each subset $S_j(j \in J) : K(S_j)$ should satisfy the inequality as below.

$$k_v \leq \frac{pt}{f} \tag{5}$$

Because we need to deploy monitoring points to achieve full coverage and guarantee timeliness, in fact, is a capacitated set cover problem which is described as follow.

$$\begin{aligned} \min\ & z = \sum_{i=1}^{m} x_i \\ & \sum_{j:e_i \in S_j} x_j \geq 1 \quad i = 1, 2, ..., n \\ s.t.\ & x_j = 0, 1 \quad i = 1, 2, ..., n \\ & |K(S_j)| \leq \tfrac{pt}{f} \end{aligned} \tag{6}$$

where $K(S_j)$ represents the number of elements of subset $S_j(j \in J)$.

4 MMCRD for Monitoring Point Deployment

Based on the above discussion, the MMCRD algorithm is summarized as Algorithm 1.

Algorithm 1. Maximum Monitoring Contribution Rate Deployment Algorithm (MMCRD)

Input:
vertex set: V=$v_1, v_2, ..., v_n$ adjacent matrix A; maximum tweet frequency: f tweets per second; data acquisition rate: p tweets per second; tolerable delay: t seconds;
Output: Monitoring Points Set MPS;
Steps:
Step 1. Initialization
Monitoring set M=Φ; Following set F=Φ;Monitored set S=Φ; Isolated set I=Φ; New Monitoring set NW=Φ;
Step 2. Determine out-degree k_i^{out} and excess average in-degree ek_i^{in} of $v_i(i = 1, 2.., n)$ and then calculate monitoring contribution rate $MCR_i(MCR_i = k_i^{out}/(ek_i^{in} + \varepsilon))$, determine the vertex v_j with maximum MCR;
Step 3. M=M+$\{v_j\}$;
Step 4. F=Φ; Find the following set of v_j and refresh F;
Step 5. Determine the size of F represented by p;
Step 6. $if p <= r = \frac{pt}{f}$,S=S+$\{v_j\}$+F;
else F=choose r vertices with lowest in-degree;
S=S+v_j+F;
end;
Step 7. Delete users from M, S and all the related edges and refresh A;
Step 8. *if* the element of A are not equal to 0,
Then jump to step 2;
else I = V-S;
end;
Step 9. While $\|\mathrm{I}\| > r$
Choose r users from I and form a set N=r users;
Create a new user v_k to follow users in set N;
NM = NM+$\{v_k\}$;
I = I-N;
end;
Step 10. Create a new user v_k to follow $\|\mathrm{I}\|$ users;
NM = NM+$\{v_k\}$;
return MPS = M+NM;

The algorithm takes a vertex set V and an adjacent matrix A of a directed graph G as inputs and return a monitoring point set MPS. The algorithm works as follows: At the beginning, we initial the monitoring set M equal to empty set, the elements of M represent the monitoring points in graph G. The elements of the monitored set S indicate the vertices have been monitored. The role of the following set F is to temporarily store the friends of a selected monitoring point. The isolated set I contains isolated nodes which are not yet monitored by monitoring points. Referred to isolated nodes, we create new vertices to cover them, these new vertices form the new monitoring set NW. In each loop, we determine the vertex v_j with maximum monitoring contribution rate (MCR) in graph G and adds it to M. Subsequently, we find the friends of v_j and temporarily store in F. If the size of F is large than timeliness constrain ($r = pt/f$), replace F

with r vertices with lowest in-degree in F, then add v_j and F to S. Loop continues until the adjacent matrix A is all 0. When the loop ends, we obtain the isolated set I by calculating the difference set of V and S. Different from step 6, for each new created vertex, we randomly choose r vertices in I to be monitored rather than choose r lowest in-degree ones for the reason that the connectivity of the remaining isolated vertices is relatively sparse and the in-degree of each vertex is very low and nearly the same. Finally, we combine the M and NW and return the monitoring point set MPS.

Figure 2 show the steps of the MMCRD algorithm using a sample graph. Figure 2(a) shows an input directed graph containing 7 vertices and 9 edges, the MCR of each vertex is initialized according to its out-degree and average in-degree of its friend(s), for example, the out-degree of C is 2, the in-degrees of its friends B and D are 2 and 4 respectively, then we can calculate the average in-degree of B and D, is equal to 3, therefore the MCR of C is 2/3. Other vertices are the same situation. The vertex with highest MCR is B, so it is added to M, both B and its friend A are added to S as shown in Fig. 2(b). B and its friend are discolored and all edges incident on them are dashed. Figure 2(c) shows E has three friends whereas up to 2 can be monitored by E such that C and F with lowest in-degree are selected. In Fig. 2(c), only two vertices D, G and an edge dg are left in graph, finally we remove them from graph and all MCRs are now 0, as shown in Fig. 2(d). The process is complete and the resultant monitoring points are (B, E, G).

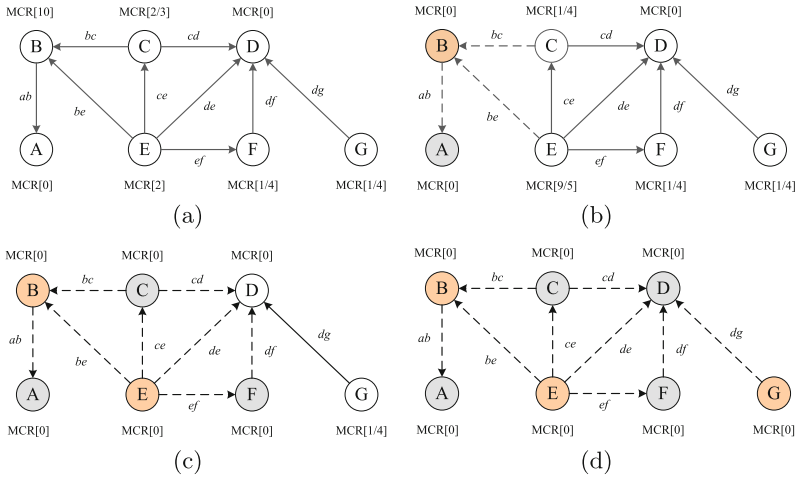
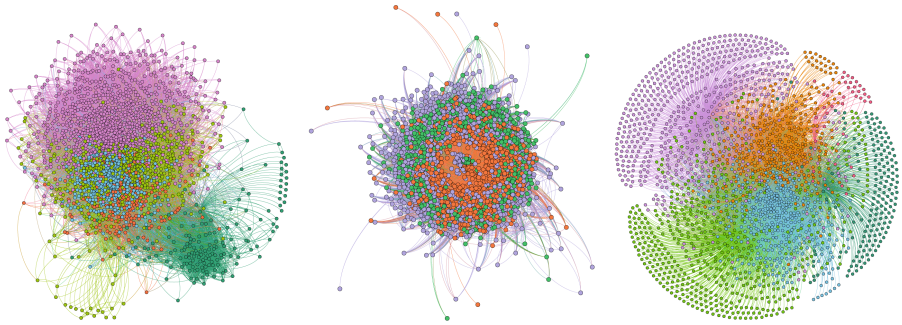


Fig. 2. Illustration of the steps of the MMCRD algorithm. (Color figure online)

5 Experiment and Evaluation

5.1 Dataset Description

In order to analyze how to deploy monitoring points on social network effectively and evaluate MMCRD algorithm, we conduct experiments on datasets derived from Facebook, Twitter, Weibo and synthetic BA scale free networks. On Facebook, Twitter and Weibo, we sample about 2000 users and their relationships relatively as research subjects. We collect all the relationships between these users. In Facebook, the number of relationships is 127841. In Twitter, the number of relationships is 182561 and 13140 relationships in Weibo. In term of synthetic network, we generate scale-free networks based on Barabasi Model [20]. We use Gephi [21], an open-source graph-visualization software to visualize the relationships network of Facebook, Twitter and Weibo respectively as shown in Fig. 3. The detailed topological characteristic of these networks as provided in Table 1.



(a) (b) (c)

Fig. 3. The relationships network. (a) Facebook relationships network. (b) Twitter relationships network. (c) Weibo relationships network.

Table 1. Topology characteristic of Facebook, Twitter, Weibo Relationships Network.

Topology characteristic	Facebook	Twitter	Weibo
Average degree	116.547	182.65	13.37
Diameter	8	5	8
Density	0.027	0.046	0.003
Modularity	0.25	0.072	0.34
Clustering coefficient	0.195	0.512	0.15
Average path length	1.977	2.088	2.857

5.2 Experiments

Based on these networks, we verify the effectiveness of MMCRD. In order to verify the effectiveness of MMCRD used in monitoring point deployment, we compare MMCRD with RD and MOD. Figure 4 shows the deployment results of RD, MOD and MMCRD in the same BA scale-free network with 20 vertices. In this case, each vertex is allowed to monitor 3 vertices at most. The vertices colored in red represent the monitoring points which are originally included in graph. The vertices colored in blue represent new created monitoring points. From the figure we can see that RD requires 9 monitors, MOD requires 8 monitors and MMCRD requires 7 monitors.

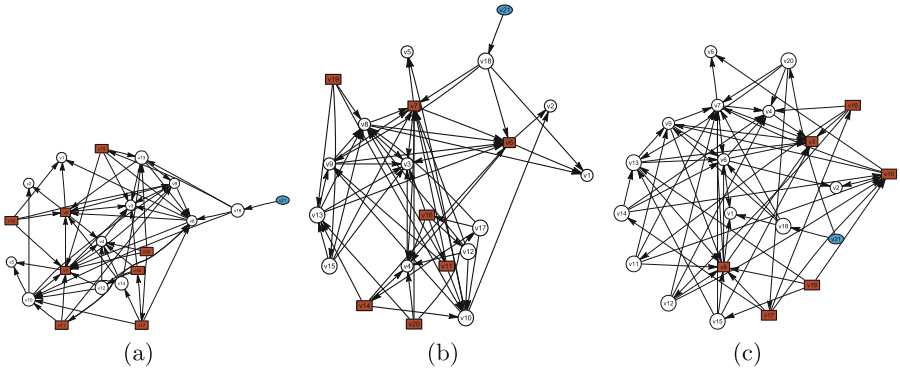


Fig. 4. Deploy monitoring points on BA scale-free network by using different deployment algorithms. (a) RD. (b) MOD. (c) MMCRD. (Color figure online)

Furthermore, we deploy monitoring points on Facebook, Twitter and Weibo and obtain the relationship between monitor capacity and required number of monitors respectively, as shown in Fig. 5. Compared to RD and MOD, given the same timeliness constraint (monitor capacity), MMCRD requires fewer monitors. We only require at most 12% of all users in order to achieve full coverage, the conclusion is coherent in Facebook, Twitter and Weibo three different relationship networks, thus we verify MMCRD is an effective approximation deployment algorithm to achieve full coverage and guarantee timeliness. For example, when the monitor capacity of each user is 50, the required number of monitors of Facebook and Weibo are both about 5% of all users and about 2.5% of all users for Twitter by using MMCRD, therefore using MMCRD to deploy monitoring points can sharply decrease the API calls in data acquisition. Furthermore, if the capacity of each monitor increase to 100, we only acquire 1% monitors in Twitter.

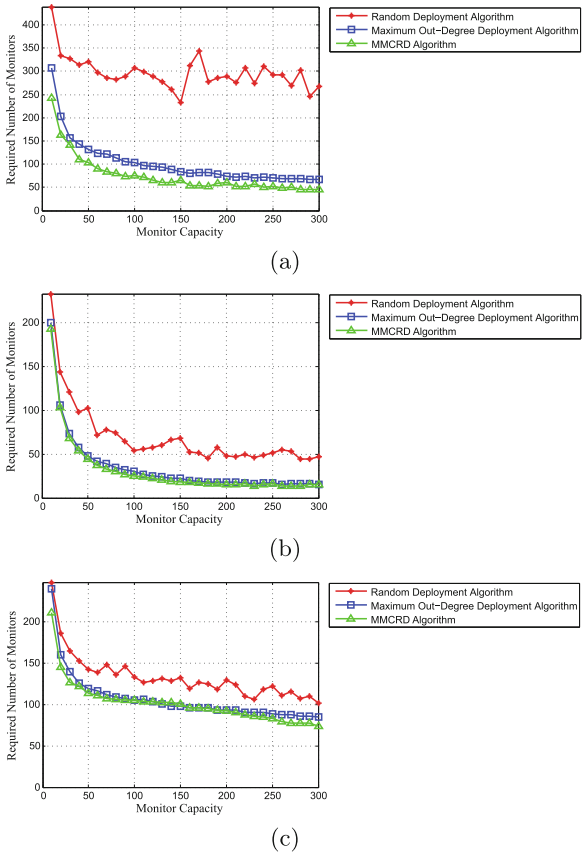


Fig. 5. The relationship between monitor capacity and the required number of monitors. (a) Facebook. (b) Twitter. (c) Weibo.

6 Conclusion and Future Work

Collecting complete and real-time social network data is the key to grasp social public opinion and mine event hot-spots. Smart deployment of monitoring points on social network is crucial in data acquisition. This paper proposed the MMCRD algorithm to deploy monitoring points on social networks based on their MCRs which is depend on out-degree and excess average in-degree of each vertex. Conducting experiment on datasets derived from real-world social networking sites and synthetic social network, we verify the effectiveness of the proposed algorithm which only require to monitor 12% of all users can achieve full coverage and guarantee timeliness. We will further consider the behaviors of different users in the network to improve our algorithm and use a more fine-grained algorithm on various social networks with different user attributes.

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